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Department of Computer Science & Engineering

Graph-Based sEMG Gesture Recognition Using NinaPro DB2
TASK 1 REPORT: Dataset Understanding, EDA, Related Work and Research Gap

Submitted To

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1. Task Overview

We began by fixing the experimental protocol before any modeling. The project uses NinaPro DB2 Exercise B (E1), covering all 40 subjects, 12 sEMG channels and 17 active gestures. For the instructor-approved repetition-wise setup, repetitions 1, 3, 4 and 6 are used on the training side, while repetitions 2 and 5 are kept untouched for later testing. Our initial checks focused on whether the data were complete, whether every gesture and repetition was represented as expected, and whether the 12 channels behaved differently enough to justify an electrode-based representation. No missing or infinite values were found in the training-side data. The raw waveforms and RMS summaries also showed clear channel- and subject-level variation. We then reviewed recent EMG-GNN work to understand how graph topology, temporal structure and generalization are handled in the literature. These observations led to the central direction of the project: compare a compact electrode graph against strong non-graph baselines under one consistent repetition-wise protocol.

2. Dataset and Experimental Protocol

Item	Observed configuration
Dataset	NinaPro Database 2 (DB2)
Exercise	Exercise B (E1)
Subjects	S1–S40 (40 subjects)
Sampling rate	2000 Hz
sEMG channels	12
Active gesture classes	17
Training repetitions	1, 3, 4, 6
Reserved test repetitions	2, 5
Window used later	200 ms = 400 samples; 50% overlap
Active training segments	2720
Segments per gesture	160
Missing / infinite values	0 / 0

3. Exploratory Data Analysis

3.1 Integrity, coverage and balance

All 40 Exercise B files were available, and every subject contained the 17 active gesture labels together with the four training repetitions used in this project. Across repetitions 1, 3, 4 and 6, we identified 2,720 active gesture-repetition segments. Each gesture contributed 160 segments and each selected repetition contributed 680. This means the original experimental design is balanced at the segment level before window generation. We also kept repetitions 2 and 5 outside the EDA so that the later test set remained untouched. Window-level counts are checked again in Task 2 because overlapping windows can introduce mild class imbalance even when the original segment counts are equal.

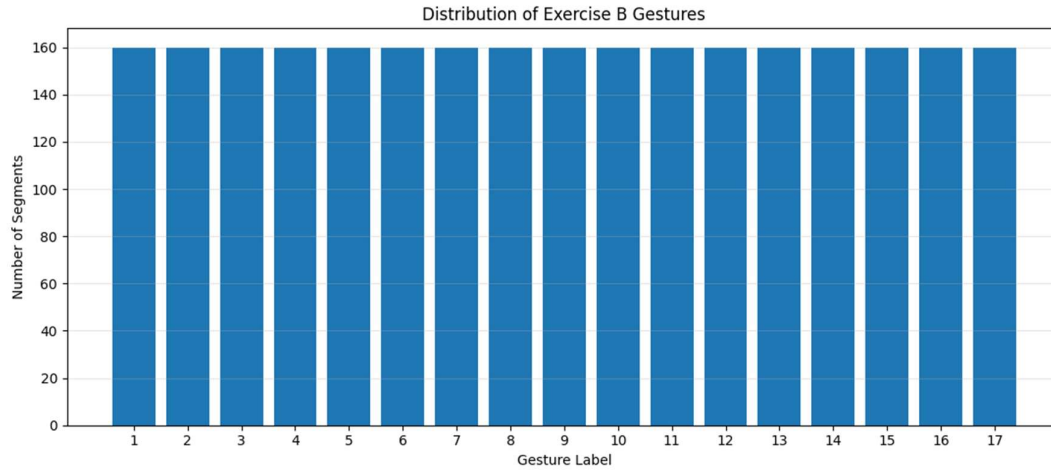


Figure 1. Distribution of Exercise B training segments across the 17 gestures.

At the segment level, every gesture contributes equally; no class is structurally under-represented before window generation.

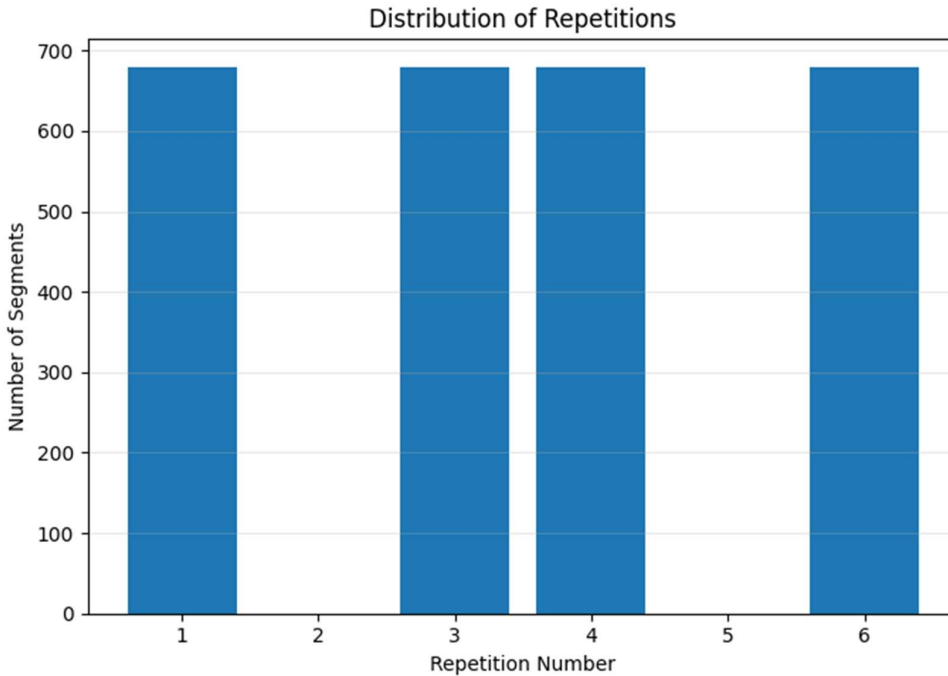


Figure 2. Distribution of the four training repetitions.

The four training repetitions contribute equally. Repetitions 2 and 5 are deliberately excluded here and reserved for the fixed test set.

3.2 Raw sEMG behavior and channel variability

The raw 12-channel signals are not interchangeable. Within the same recording interval, some electrodes show stronger amplitudes or sharper transient activity than others. The RMS summaries tell the same story: Channel 8 has the highest median RMS across subjects, while Channel 11 has the lowest. We also observe visible subject-to-subject variation in median RMS. These differences are important because they support keeping electrode identity instead of flattening all channels into one undifferentiated vector.

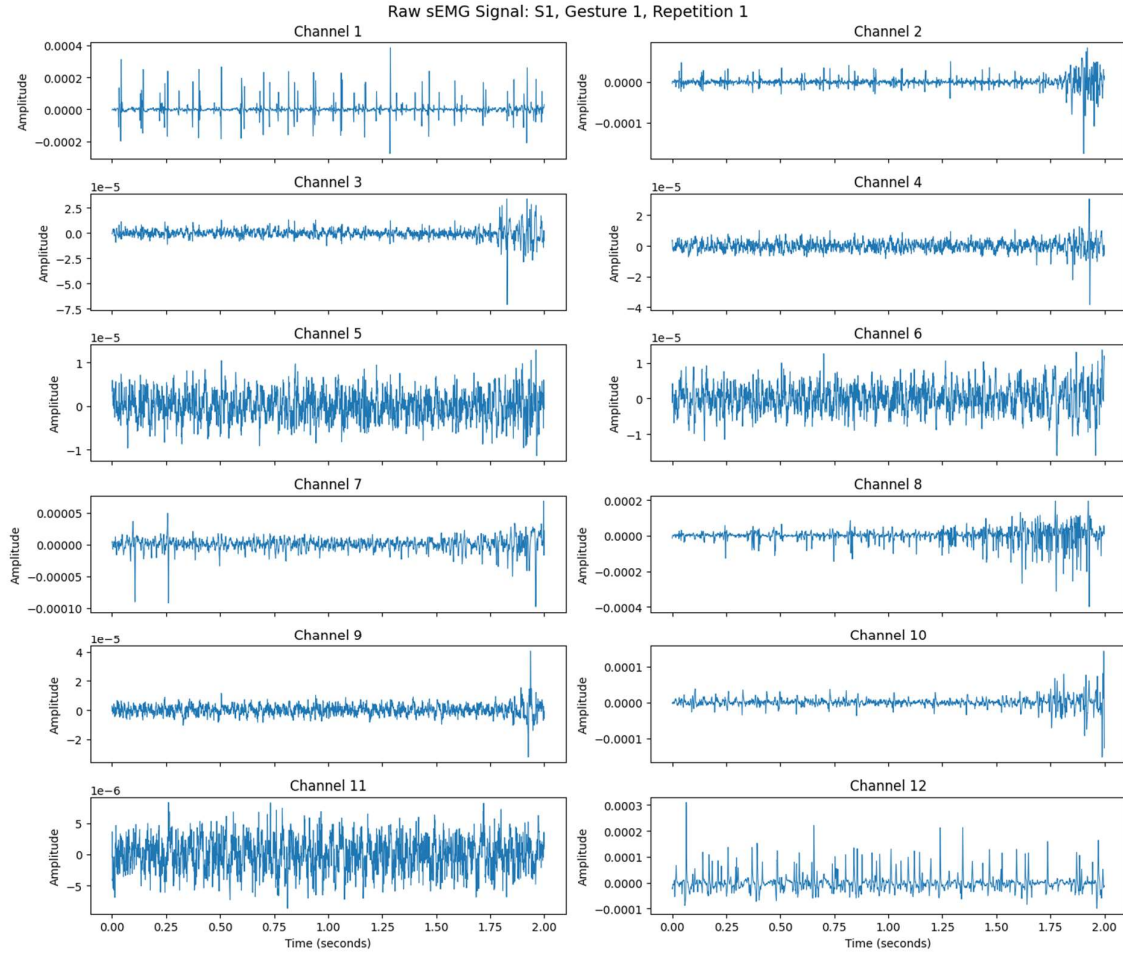


Figure 3. Example 12-channel raw sEMG waveform from Exercise B.

The 12 channels show visibly different amplitudes and transient patterns within the same interval, supporting per-electrode feature extraction.

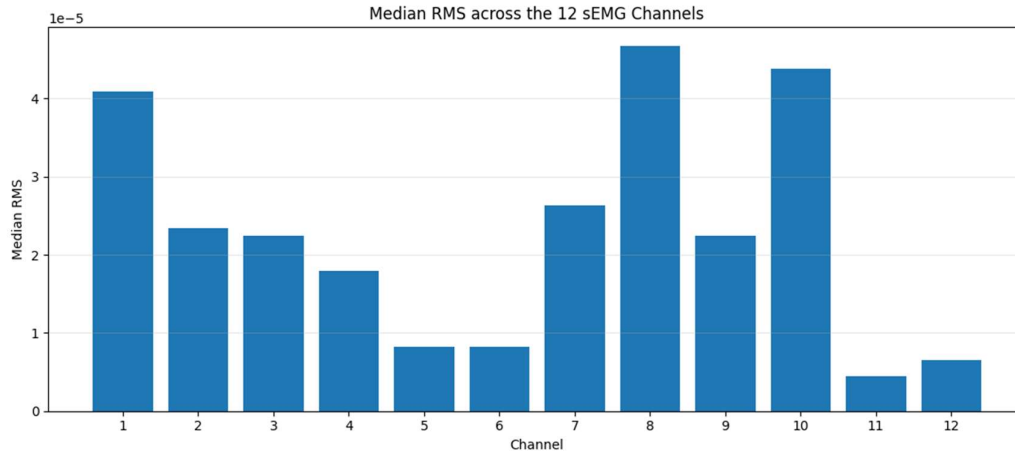


Figure 4. Median RMS across the 12 sEMG channels.

RMS energy is not uniform across channels, which gives a practical reason to preserve electrode identity in the later graph model.

4. Related Work

The literature review was used to answer a practical question: when does a graph add information that a strong feature-based classifier does not already capture? The selected studies cover adaptive or spatiotemporal graph construction, cross-user generalization, real-time deployment and hybrid CNN-GCN learning. Because the datasets and evaluation protocols differ, their reported accuracies are treated as context rather than as directly comparable

targets. The full paper-by-paper comparison is kept in the separate related-work deliverable; this report highlights the findings that directly shaped our model design.

Study	Dataset	Graph/model idea	Result and relevance
Zhou et al. (2025)	NinaPro DB2 + UOW	CovGCN with covariance-based adaptive topology and attention	Exact DB2 GNN reference; accessible abstract reports improvement over ML baselines but does not provide a directly usable exact score.
Lee et al. (2025)	DB5, BandMyo, UCI, CapgMyo	Spatial / spatiotemporal GCN with KNN or fully connected graphs and temporal edges	92.4% inter-subject accuracy on DB5 for one graph setting; demonstrates graph-topology sensitivity.
Xu et al. (2024)	HD-sEMG, 11 subjects, 17 gestures	CNN-MSTGCN / TL-CMSTGCN	68.0% user-independent and 92.3% after target-user transfer learning; highlights subject variability.
Vipulanandan et al. (2024)	Private MyoBand, 8 electrodes	Real-time muscle-activation GNN	99% average accuracy with ~48 ms graph construction + prediction; protocol and private dataset limit direct comparison.
Xu & Jiang (2023)	NinaPro DB1	Hybrid ResNet + GCN joint voting	90.07% accuracy on DB1; shows complementary value of graph-derived muscle-synergy information.

5. Research Gap and Proposed Direction

1. Exact graph-based evidence on NinaPro DB2 is still limited, especially when the gesture subset and evaluation protocol are held fixed.
2. Graph construction is not standardized. Recent studies use KNN, fully connected, covariance-based, temporal or adaptive edges, so the value of connectivity itself is difficult to isolate.
3. Several published models change the CNN, attention mechanism and graph at the same time. A controlled no-graph and edge-weight ablation is therefore needed to see whether the graph itself helps.
4. A graph model should also be compared with a strong classical baseline. For engineered EMG features, XGBoost and Random Forest are difficult benchmarks and should not be omitted.
5. Finally, accuracy alone is not enough. Stability across folds, statistical testing and interpretable evidence are needed if the graph contribution is to be convincing.

Working research question. Can a compact electrode graph built from the 12 NinaPro DB2 sEMG channels improve 17-class Exercise B recognition over strong non-graph baselines under the same repetition-wise protocol, and can that improvement be supported by controlled ablation, cross-validation and interpretable evidence?

6. Conclusion

Task 1 gave us a clear starting point rather than a performance claim. The Exercise B data are complete on the training side, balanced at the gesture-segment level and visibly heterogeneous across electrodes and subjects. The literature also shows that graph design must be tested rather than assumed. For that reason, Task 2 moves to a strong baseline suite and an electrode-aware ST-EGAT model, while Task 3 later checks whether the graph still helps under controlled ablation, grouped cross-validation and explainability.

References

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